

SAINT-MALO-DINARD AP, France

WMO: 071250

Lat: 48.5889N

Lon: 2.0748W

Elev: 219

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	30.6	21.1	16.0	31.1	24.5	18.8	33.2	30.4	50.3	27.8	50.6	6.2	90	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	80.4	67.3	76.3	65.5	73.3	63.9	68.5	77.5	66.7	74.0	65.2	71.4	9.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.2	72.7	63.8	89.3	70.1	62.5	85.3	68.5	32.8	77.7	31.4	74.4	30.1	71.4	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
24.9	21.8	19.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			22.5	89.5	3.0	5.2	20.3	93.2	18.5	96.3	16.8	99.2	14.6	103.0
			21.7	72.4	3.1	2.6	19.4	74.3	17.6	75.9	15.8	77.4	13.5	79.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	43.5	43.8	46.9	50.4	55.4	60.6	63.6	63.4	60.8	55.6	49.0	44.5	(6)
(7)		DBStd	8.94	6.15	5.65	5.25	4.86	4.50	4.74	3.97	3.96	4.51	5.06	4.92	6.17	(7)
(8)		HDD50	844	211	181	124	51	6	0	0	0	11	75	185		(8)
(9)		HDD65	4407	666	593	560	438	300	150	73	77	140	295	481	634	(9)
(10)		CDD50	2004	9	9	29	63	172	317	421	414	325	184	45	16	(10)
(11)		CDD65	91	0	0	0	0	2	16	30	26	15	2	0	0	(11)
(12)		CDH74	614	0	0	0	4	25	142	194	155	87	7	0	0	(12)
(13)		CDH80	141	0	0	0	0	3	38	47	38	14	1	0	0	(13)
(14)	Wind	WSAvg	9.8	11.1	11.1	10.4	9.8	9.4	8.8	8.9	8.6	8.6	9.3	10.5	10.9	(14)
(15)	Precipitation	PrecAvg	28.5	2.6	2.3	1.8	2.1	2.1	1.9	1.9	2.1	2.2	3.0	3.5	3.1	(15)
(16)		PrecMax	34.5	5.2	5.1	5.6	5.0	4.7	4.0	3.9	5.1	5.8	6.0	6.6	7.1	(16)
(17)		PrecMin	21.7	0.3	0.4	0.4	0.3	0.5	0.5	0.7	0.4	0.3	0.6	0.9	0.6	(17)
(18)		PrecStd	3.7	1.5	1.2	1.2	1.1	1.0	1.0	0.9	1.3	1.2	1.3	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	59.4	66.3	73.3	77.4	85.5	85.6	85.8	82.1	72.5	63.5	58.5	(19)
(20)			MCWB	54.3	52.7	55.1	57.9	65.2	70.5	69.2	70.3	66.8	63.6	59.5	55.3	(20)
(21)		2%	DB	55.9	56.0	61.5	67.0	71.0	77.6	78.9	78.1	75.9	67.9	60.5	56.8	(21)
(22)			MCWB	52.9	51.4	53.0	56.9	61.1	66.3	66.4	66.6	65.2	61.6	57.0	54.1	(22)
(23)		5%	DB	53.9	54.0	58.0	62.5	67.3	73.1	75.0	74.0	72.0	65.1	58.2	55.2	(23)
(24)			MCWB	51.2	50.4	52.1	54.6	59.1	63.8	64.7	64.9	63.4	60.0	54.9	52.7	(24)
(25)		10%	DB	52.1	52.2	55.4	58.8	64.1	69.7	72.0	71.2	69.2	63.0	56.4	53.5	(25)
(26)			MCWB	49.6	49.0	50.9	52.2	57.4	61.7	63.5	63.1	62.0	59.0	53.5	51.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	54.3	56.7	61.1	66.0	71.6	71.1	70.6	69.4	65.1	60.6	56.1	(27)
(28)			MCDWB	57.4	58.0	63.3	69.4	76.0	80.6	83.4	83.2	77.9	70.1	62.7	57.9	(28)
(29)		2%	WB	53.5	52.6	54.5	57.5	62.1	67.4	68.0	67.8	66.5	62.7	57.4	54.6	(29)
(30)			MCDWB	55.7	55.3	59.2	65.6	69.1	76.4	76.2	75.4	73.3	66.4	59.9	56.4	(30)
(31)		5%	WB	51.7	51.0	53.0	55.2	59.8	64.6	66.0	65.9	64.6	61.1	55.6	53.1	(31)
(32)			MCDWB	53.5	53.5	57.2	61.4	65.9	71.5	72.7	72.1	70.2	64.2	57.8	55.0	(32)
(33)		10%	WB	50.0	49.3	51.3	53.1	58.0	62.3	64.4	64.4	62.9	59.5	53.9	51.2	(33)
(34)			MCDWB	51.9	51.9	55.1	57.8	63.0	68.3	70.5	69.7	67.7	62.3	56.0	53.2	(34)
(35)	Mean Daily Temperature Range		MDBR	9.5	11.0	12.9	14.7	14.6	15.3	15.4	15.3	15.6	12.5	10.4	9.5	(35)
(36)		5% DB	MCDBR	9.8	12.6	17.1	21.3	20.6	21.8	21.5	20.4	19.6	13.9	11.6	9.1	(36)
(37)			MCWBR	8.8	9.3	10.7	12.3	12.0	12.0	11.1	10.3	10.5	8.6	8.9	8.2	(37)
(38)		5% WB	MCDBR	9.4	11.5	15.2	18.9	18.5	19.9	18.8	17.7	17.2	12.6	10.8	9.2	(38)
(39)			MCWBR	8.9	9.4	10.2	11.7	11.5	11.8	10.8	10.2	10.1	8.7	9.2	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.345	0.374	0.389	0.395	0.392	0.389	0.380	0.365	0.358	0.334	0.322	(40)	
(41)		taud	2.424	2.394	2.314	2.288	2.315	2.356	2.365	2.413	2.434	2.449	2.442	2.396	(41)	
(42)		Ebn at Noon	232	255	265	272	275	276	275	273	266	247	227	216	(42)	
(43)		Edh at Noon	21	27	34	38	39	38	37	34	30	25	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	306	554	897	1376	1630	1798	1718	1466	1158	667	374	258	(44)	
(45)		RadStd	28	64	98	142	137	196	111	120	87	54	38	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page